Inner Product Spaces

Problem sheet for the Day 8 class · CSIR NET / GATE previous year questions

Name: Date: Score: / 10

Instructions. 10 problems from CSIR NET, GATE (MA) and IIT JAM. A tag with an exam and year marks an actual previous-year question; a tag reading *Exam pattern* marks a problem written in the same style. MCQ: one correct option. MSQ: one or more correct options. NAT: write the number. Suggested time: 30 minutes. **The answer key is posted 24 hours after this sheet.**

1. GATE 2013 MCQ The possible set of eigen values of a 4×4 skew-symmetric orthogonal real matrix is

(A) $\{\pm i\}$
(B) $\{\pm i, \pm 1\}$
(C) $\{\pm 1\}$
(D) $\{0, \pm i\}$

2. CSIR NET Dec 2016 MCQ The matrix $\begin{pmatrix} 3 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 3 \end{pmatrix}$ is

(A) positive definite.
(B) non-negative definite but not positive definite.
(C) negative definite.
(D) neither negative definite nor positive definite.

3. GATE 2013 MCQ Let B be a real symmetric positive-definite $n \times n$ matrix. Consider the inner product on $\mathbb{R}^n$ defined by $\langle X, Y \rangle = Y^t B X$. Let A be an $n \times n$ real matrix and let $T : \mathbb{R}^n \to \mathbb{R}^n$ be the linear operator defined by $T(X) = AX$ for all $X \in \mathbb{R}^n$. If S is the adjoint of T, then $S(X) = CX$ for all $X \in \mathbb{R}^n$, where C is the matrix

(A) $B^{-1} A^t B$
(B) $B A^t B^{-1}$
(C) $B^{-1} A B$
(D) A^t

4. GATE 2015 MCQ Let V be a closed subspace of $L^2[0,1]$ and let $f, g \in L^2[0,1]$ be given by $f(x) = x$ and $g(x) = x^2$. If $V^\perp = \text{Span}\{f\}$ and Pg is the orthogonal projection of g on V, then $(g - Pg)(x)$, $x \in [0,1]$, is

(A) $\frac{3}{4}x$
(B) $\frac{1}{4}x$
(C) $\frac{3}{4}x^2$
(D) $\frac{1}{4}x^2$

5. GATE 2015 MCQ Let M be an invertible Hermitian matrix and let $x, y \in \mathbb{R}$ be such that $x^2 < 4y$. Then

(A) both $M^2 + xM + yI$ and $M^2 - xM + yI$ are singular
(B) $M^2 + xM + yI$ is singular but $M^2 - xM + yI$ is non-singular
(C) $M^2 + xM + yI$ is non-singular but $M^2 - xM + yI$ is singular
(D) both $M^2 + xM + yI$ and $M^2 - xM + yI$ are non-singular

6. GATE 2015 NAT Let $W = \text{Span}\{\frac{1}{\sqrt{2}}(0,0,1,1),\ \frac{1}{\sqrt{2}}(1,-1,0,0)\}$ be a subspace of the Euclidean space $\mathbb{R}^4$. Then the square of the distance from the point $(1,1,1,1)$ to the subspace W is equal to _____.

7. CSIR NET June 2016 MSQ Suppose $\{v_1, \ldots, v_n\}$ are unit vectors in $\mathbb{R}^n$ such that $\|v\|^2 = \sum_{i=1}^{n} |\langle v_i, v\rangle|^2$ for all $v \in \mathbb{R}^n$. Then decide the correct statements in the following:

(A) $v_1, \ldots, v_n$ are mutually orthogonal.

(B) $\{v_1, \ldots, v_n\}$ is a basis for $\mathbb{R}^n$.

(C) $v_1, \ldots, v_n$ are not mutually orthogonal.

(D) At most $n-1$ of the elements in the set $\{v_1, \ldots, v_n\}$ can be orthogonal.

8. CSIR NET June 2016 MSQ Let $V = \{f : [0,1] \to \mathbb{R} \mid f \text{ is a polynomial of degree less than or equal to } n\}$. Let $f_j(x) = x^j$ for $0 \le j \le n$ and let A be the $(n+1) \times (n+1)$ matrix given by $a_{ij} = \int_0^1 f_i(x) f_j(x)\, dx$. Then which of the following is/are true?

(A) $\dim V = n$.

(B) $\dim V > n$.

(C) A is nonnegative definite, i.e., for all $v \in \mathbb{R}^n$, $\langle Av, v\rangle \ge 0$.

(D) $\det A > 0$.

9. CSIR NET Dec 2017 MSQ For every 4×4 real symmetric non-singular matrix A, there exists a positive integer p such that

(A) $pI + A$ is positive definite

(B) A^p is positive definite

(C) A^{-p} is positive definite

(D) $\exp(pA) - I$ is positive definite

10. CSIR NET Dec 2017 MSQ Let A be an $m \times n$ matrix of rank m with $n > m$. If for some non-zero real number α, we have $x^t A A^t x = \alpha x^t x$ for all $x \in \mathbb{R}^m$, then $A^t A$ has

(A) exactly two distinct eigenvalues

(B) 0 as an eigenvalue with multiplicity $n - m$

(C) α as a non-zero eigenvalue

(D) exactly two non-zero distinct eigenvalues